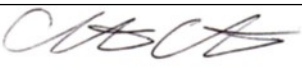


Guideline Title: **Cardioversion**

Guideline Number: **GUID-3203-PR004**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To restore hemodynamic stability in a patient who exhibits a tachydysrhythmia that compromises effective perfusion
- II. **DEPARTMENT GUIDELINE:**
 - A. **INDICATIONS:**
 - i. Ventricular tachycardia with a pulse (unstable)
 - ii. Atrial fibrillation with rapid ventricular response (unstable)
 - iii. Atrial flutter with rapid ventricular response (unstable)
 - iv. Supraventricular tachycardia (unstable)
 - B. **CONTRAINDICATIONS:** Patient is wet or for any other reason cannot be electrically isolated from the medical team members.
 - C. **EQUIPMENT:**
 - i. Cardiac monitor with ability to cardiovert
 - ii. Conductive hands-free defibrillation pads
 1. Adult
 2. Pediatric pads for infants/children under 15 kg.
 - D. **PROCEDURE:**
 - i. If the procedure is to be performed in flight, notify the pilot of the intent to cardiovert. Do NOT proceed further until authorized to do so by the pilot.
 - ii. Place conductive hands-free pads on the patient's bare chest.
 1. Anterior / Posterior: place negative electrode on left anterior chest, halfway between the xiphoid process and the left nipple. This corresponds with the V2-V3 ECG position. Place the positive electrode on left posterior chest beneath the scapula and lateral to the spine.
 2. Anterior / Anterior: place the negative electrode on the left chest, mid-axilla over the fourth intercostal space. Place positive electrode on anterior right chest, sub-clavicular area.
 3. Avoid placing the pads over any implantable electrical device.

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- iii. Confirm the need to cardiovert per *GUID3203-C003 Atrial Fibrillation & Atrial Flutter or GUID3203-C006 Narrow and Wide Complex Tachycardia*
- iv. Consider sedation or analgesia per *GUID3203-G005 Analgesia and Sedation Management* if MAP > 65 mmHg.
- v. Turn the monitor/defibrillator on. Locate defib mode if using a non-department monitor.
- vi. Initiate the “synchronize” mode by pressing SYNC on the Zoll X Series and confirm that indicator markers appear on the oscilloscope on the R waves of each complex (the size of the complex may need to be adjusted or the lead may need to be changed for adequate synch confirmation).
- vii. Select the joule dose by adjusting the energy level arrow buttons based on current recommended dosing.
 - 1. For Adult:
 - a. For narrow regular rhythm: 50-100 J
 - b. For narrow irregular rhythm: 120-200 J
 - c. For wide regular complex: 100 J
 - d. For wide irregular: Defibrillation dose, not synchronized
 - e. If no response after initial cardioversion, give 2nd dose of 200 J.
 - f. Subsequent doses should be higher than initial dose up to max joule dose of monitor. (Other facility/agency monitors maximum may be up to 360 J).
 - 2. For Pediatric:
 - a. Initial: Synchronized cardiovert at 0.5 to 1 J/kg.
 - b. Subsequent doses: Synchronized cardiovert at 2 J/kg.
 - c. If no response to therapy, contact Medical Control.
- viii. Charge the monitor/defibrillator to the appropriate energy level.
- ix. When defibrillator is fully charged, notify team members of the intent to cardiovert by stating loudly and clearly “CLEAR for shock” and assuring that nobody is touching the patient or stretcher.
- x. Push the SHOCK button to discharge the shock, realizing that it may take a few seconds to synchronize to the rhythm and cardiovert the patient.
- xi. If the synchronization is not coordinating and the patient is markedly unstable, unsynchronized cardioversion should be performed by inactivating the “synchronize” mode and performing defibrillation at the appropriate energy level.
- xii. Observe for any change in rhythm or vital signs.
- xiii. Repeat as appropriate for the patient’s condition ensuring the sync mode is selected.

III. DOCUMENTATION:

- A. EMS charts

IV. REFERENCES:

- A. American Heart Association, 2015.